Set operations: If A and B are any two sets, then

(i) Union of two sets: $A \cup B = \{x: x \in A \text{ or } x \in B\}$

(ii) Intersection of two sets: $A \cap B = \{x: x \in A \,\&\, x \in B\}$

(iii) Complement of a set: A' or $\bar{A} = \{x: x \notin A\}$

(iv) Difference of two sets: $A - B = \{x: x \in A \text{ but } x \notin B\}$

Commutative law: $A \cup B = B \cup A$ and $A \cap B = B \cap A$

Associative law: $(A \cup B) \cup C = A \cup (B \cup C)$ and $(A \cap B) \cap C = A \cap (B \cap C)$

Distributive law: $A \cup (B \cap C) = (A \cup B) \cap (A \cup C)$ and $A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$

Complementary law: $A \cup A' = S$ and $A \cap A' = \emptyset$

De Morgan's laws: $(\boldsymbol{A} \cup \boldsymbol{B})' = \boldsymbol{A}' \cap \boldsymbol{B}'$ and $(\boldsymbol{A} \cap \boldsymbol{B})' = \boldsymbol{A}' \cup \boldsymbol{B}'$

Probability helps to determine the likelihood of events. If E is an event of a sample space S for an experiment, then we have the following:

$$P(E) = \frac{\text{Number of favourable outcomes to happen } A}{\text{Total number of possible outcomes of the experiment}} = \frac{n(E)}{n(S)}$$

Axiomatic approach: It is a rule that associates with each event a real number $P(E)$ which satisfies the following three axioms.

Axiom 1: For any event E, $0 \leq P(E) < 1$

Axiom 2: $P(S) = 1$

Axiom 3: If $E_1, E_2, \dots, E_n$ are a finite number of disjoint events ($E_i \cap E_j = \emptyset\,; i \neq j$) of S, then

$$P(E_1 \cup E_2 \cup \dots \cup E_n) = P(E_1) + P(E_2) + \cdots + P(E_n) = \sum P(E_i)$$

Theorems on probability:

Theorem 1: Probability of an impossible event is zero. i.e., $P(\emptyset) = 0$.

Theorem 2: Probability of the complementary event A' or $\bar{A}$ of A is given by, $P(A') = 1 - P(A)$.

Theorem 3: For any two events A and B, $P(A' \cap B) = P(B) - P(A \cap B)$.

Theorem 4: For any two events A and B, $P\,(A) = P(A \cup B') - P(A \cup B)'$

Theorem 5: If A and B are two events such that $A \subset B$, then $P(B \cap A') = P(B) - P(A)$.

Theorem 6: If $B \subset A$, then $P(B) \leq P(A)$.

Theorem 7: If $A \cap B = \emptyset$, then $A \subset B'$ and hence $P(A) \leq P(B')$.

Law of addition for probability:

(i) $P(A \cup B) = P(A) + P(B) - P(A \cap B)$

(ii) $P(A \cup B \cup C) = P(A) + P(B) + P(C) - P(A \cap B) - P(A \cap C) - P(B \cap C) + P(A \cap B \cap C)$

For mutually exclusive or disjoint events

(i) $P(A \cup B) = P(A) + P(B)$,

(ii) $P(A \cup B \cup C) = P(A) + P(B) + P(C)$

Conditional probability:
Conditional probability refers to the likelihood of an event happening, given that another event has already occurred. It is essentially the probability of an event E_r happening, knowing that the event E_c has already happened. This is represented mathematically as $P(E_r|E_c)$, read as "the probability of E_r given E_c. Conditional probability is particularly useful when dealing with dependent events, meaning the occurrence of one event affects the probability of the other.

Formulae:

(i) The conditional probability of an event A, given that event B has occurred, is defined by

$$P(A|B) = \frac{n(A \cap B)}{n(B)} = \frac{\frac{n(A \cap B)}{n(S)}}{\frac{n(B)}{n(S)}} = \frac{P(A \cap B)}{P(B)}, \text{ provided that } n(B) > 0$$

(ii) The conditional probability of an event B, given that event A has occurred, is defined by

$$P(B|A) = \frac{n(A \cap B)}{n(A)} = \frac{\frac{n(A \cap B)}{n(S)}}{\frac{n(A)}{n(S)}} = \frac{P(A \cap B)}{P(A)}, \text{ provided that } n(A) > 0$$

(iii) The probability that two events, A and B, both occur is given by the multiplication rule,

$$P(A \cap B) = P(B)P(A|B) \text{ or } P(A \cap B) = P(A)P(B|A)$$

For the complementary issue

$$P(A'|B) = 1 - P(A|B) \text{ and } P(B'|A) = 1 - P(B|A)$$

For $A \subseteq B \rightarrow A \cap B = A$, we get

(i) The conditional probability of an event A, given that event B has occurred, is defined by

$$P(A|B) = \frac{n(A \cap B)}{n(B)} = \frac{n(A)}{n(B)} = \frac{P(A)}{P(B)}$$

(ii) The conditional probability of an event B, given that event A has occurred, is defined by

$$P(B|A) = \frac{n(A \cap B)}{n(A)} = \frac{n(A)}{n(A)} = 1$$

For $B \subseteq A \rightarrow A \cap B = B$, we get

(i) The conditional probability of an event A, given that event B has occurred, is defined by

$$P(A|B) = \frac{n(A \cap B)}{n(B)} = \frac{n(B)}{n(B)} = 1$$

(ii) The conditional probability of an event B, given that event A has occurred, is defined by

$$P(B|A) = \frac{n(A \cap B)}{n(A)} = \frac{n(B)}{n(A)} = \frac{P(B)}{P(A)}$$

For $A \cap B = \emptyset \rightarrow n(A \cap B) = 0$, we get

$$P(A|B) = 0 = P(B|A)$$

Note that, $P(A|A) = 1$ and $P(B'|B) = 0$

Extensions of the conditional probability

$$P(A|B_1 \cup B_2 \cup \cdots \cup B_m) = \frac{n(A \cap B_1) + n(A \cap B_2) + \cdots + n(A \cap B_m)}{n(B_1) + n(B_2) + \cdots + n(B_m)}$$
$$= \frac{P(A \cap B_1) + P(A \cap B_2) + \cdots + P(A \cap B_m)}{P(B_1) + P(B_2) + \cdots + P(B_m)}$$

$$P(A_1 \cup A_2 \cup \cdots \cup A_m|B) = \frac{n(A_1 \cap B) + n(A_2 \cap B) + \cdots + n(A_m \cap B)}{n(B)}$$
$$= \frac{P(A_1 \cap B) + P(A_2 \cap B) + \cdots + P(A_m \cap B)}{P(B)}$$

Sequence of conditional events

$$P(A \cap B \cap C \cap D) = P(A)P(B|A)P(C|A \cap B)P(D|A \cap B \cap C)$$

Independent events:

Independent events are events where the occurrence of one does not affect the probability of the other. If the events A and B are independent, then $P(A|B) = P(A)$ and $P(B|A) = P(B)$.

Then, we get

$$P(A \cap B) = P(A)P(B|A) = P(A) \times P(B) = P(A)P(B)$$
$$P(A \cap B) = P(B)P(A|B) = P(B) \times P(A) = P(B)P(A)$$

If A and B are independent A and B', A' and B, and A' and B' are independent as well. Because of that

$$P(A \cap B') = P(A)P(B')$$
$$P(A' \cap B) = P(A')P(B)$$
$$P(A' \cap B') = P(A')P(B')$$

Mutual Independence: The events A, B, and C are mutually independent if and only if the following two conditions are satisfied:

(i) $P(A \cap B) = P(A)P(B)$, $P(B \cap C) = P(B)P(C)$, and $P(A \cap C) = P(A)P(C)$.

(ii) $P(A \cap B \cap C) = P(A)P(B)P(C)$.

If some of the events are mutually independent, all of their combinations of complements and set operations are mutually independent according to their fundamental laws.

Bayes' theorem:

Bayes' theorem is a mathematical formula used in probability theory to estimate unknown conditional probabilities based on the known conditional probabilities. It describes how likely an event is based on prior knowledge of conditions that might be related to the event. In essence, it helps update the probability of a hypothesis (event) given new evidence.

Analogy of Bayes' theorem:
Bayes' Theorem allows estimating the posterior probability (the updated belief) by combining the prior probability (the initial belief) with the likelihood of observing the new evidence, given that the hypothesis is true.

Formula: The general formula for Bayes' theorem is:

$$P(A_k|B) = \frac{P(A_k \cap B)}{P(B)} = \frac{P(B|A_k)P(A_k)}{P(B|A_1)P(A_1) + P(B|A_2)P(A_2) + \cdots + P(B|A_n)P(A_n)}$$

where,

(i) The event A_k is an event among $A_i; 1 \le i \le n$ that has to be aimed for a certain requirement of practical purposes.

(ii) $P(A_k|B)$ is called the posterior probability and describes the probability of an event A_k occurring, given that event B has occurred.

(iii) $P(B|A_i)$ is called the likelihood and describes the probability of event B occurring, given that event A_i has occurred.

(iv) $P(A_i)$ is the prior probability of the event A_i occurring.

(v) $P(B)$ is the prior probability of event B occurring, which is also called "the law of total probability".